Glutathione

SC 200–400 mg daily OR 300–600 mg every other day

Glutathione (GSH) is the endogenous tripeptide γ-glutamyl-cysteinyl-glycine. [3,16] Glutathione is investigated for antioxidant support, liver-injury markers, skin pigmentation, exercise-related fatigue, and immune-cell function. [3,4,8–11] It participates in peroxide reduction, conjugation reactions, and protein-thiol redox regulation. [3] IV administration transiently raises circulating glutathione and alters thiol metabolites. [4,6,7,18] Subcutaneous (SC) wellness use and the doses discussed here remain experimental extrapolations.

💉 SC tolerability is a plausible extrapolation from selected short-term IV findings. [5,13,14]

⭐ SC benefits are plausible extrapolations from oral, IV, and mechanistic findings. [3–11]

Caution: Cancer or anticancer treatment warrants review with the treating oncology team. [3,15]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

In a 21-person Parkinson disease trial, IV glutathione given three times weekly for four weeks had adverse-event reporting similar to placebo and no adverse-event withdrawals. The NAFLD pilot recorded a small HbA1c rise and withdrawals for fatigue, elevated blood pressure, and rash; its uncontrolled design limits causal interpretation.

Dosage & Patterns

An SC depot could spread absorption over time and lower peak exposure relative to an IV bolus. Repeated administration could replenish circulating glutathione or its metabolites. These are mechanistic extrapolations from IV handling; the interval depends on SC absorption, degradation, and tissue handling. [6,7]

Human study regimens

- **Oral: 250–1,000 mg/day** in studies of glutathione stores, liver markers, skin pigmentation, or exercise-related fatigue. Treatment durations and populations differed. [2,4,8–11]
- **IV: 1,400 mg three times weekly for four weeks** in a small Parkinson disease trial; it did not show a significant symptomatic advantage over placebo. [5] The studied 1,400 mg IV three-times-weekly regimen totals 4,200 mg/week, averaging 600 mg/day. [5] This arithmetic does not establish equivalent SC exposure, efficacy, or tolerability.

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Pharmacy context

Strive Pharmacy has published 200 mg SC one to two times weekly or 100 mg SC every other day as practice examples, lower than the community schedules above. These are pharmacy schedules rather than clinical regimens. [17]

The following speculative, experimental SC doses and patterns are discussed in peptide communities:

- **200–400 mg SC once daily**
- **300–600 mg SC every other day**

Benefits

- **Plausible improvement in liver-injury markers** in NAFLD. In an uncontrolled pilot, 29 completers had lower ALT after 300 mg/day orally for four months, following a three-month lifestyle intervention. Overall liver-fat and stiffness measures did not change significantly. [8]
- **Plausible support for glutathione-dependent conjugation when glutathione availability is limiting**. Endogenous GSH participates in enzyme-mediated conjugation of electrophilic compounds. [3] Supplemental benefit depends on whether it increases GSH where needed; enhanced toxin clearance is a mechanistic extrapolation rather than a measured outcome in the cited supplementation trials. [1,2,4]
- **Plausible modest skin lightening** with oral supplementation. One 60-person trial of 500 mg/day orally for four weeks found a greater reduction in melanin index than placebo at two of six sites. [9] In another trial using 250 mg/day for 12 weeks, overall pigmentation comparisons were not significant, although an age-subgroup result at one site favored reduced GSH. [10]
- **Plausible reduction in exercise-related fatigue** with oral use. An eight-man crossover trial using 1 g/day orally for two weeks reported favorable fatigue-related mood scores and lower perceived exertion at the end of a cycling bout. Mouse findings included higher muscle PGC-1α, AMPK protein, and mitochondrial DNA and less exercise-associated interstitial acidification. [11] These results support a mechanistic hypothesis of altered muscle metabolism; general recovery and chronic-fatigue applications remain speculative.
- **Increased natural-killer-cell activity** in one oral trial. The high-dose arm of the six-month oral trial showed increased natural-killer-cell cytotoxicity at three months in a tested subset. [4] This laboratory finding supports a plausible immune effect; protection against infections is an extrapolated clinical endpoint.

Duration for Functional Benefits

IV concentrations fall rapidly. The duration of SC symptom benefits cannot be estimated from these IV measurements. [6,7]

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Applications

Studied uses include skin pigmentation, exercise-related fatigue, and liver markers in NAFLD. [8–11,19]

Cautions

Cancer and anticancer therapy: Effects on tumor redox biology and treatment response are context-dependent. [3] A 50-patient randomized gastric-cancer trial found less cisplatin-associated neuropathy with a supervised IV-plus-IM glutathione regimen and did not observe reduced tumor response in that trial. [15] Use alongside cancer treatment should be assessed by the treating oncology team for the specific regimen.

Side Effects

Hypersensitivity, including anaphylaxis: Serious reactions have been reported with IV administration. [13] Breathing difficulty, fainting, or signs of shock after injection require emergency care. [13,14]

Drivers of Diminished Response

Receptor desensitization is not the principal expected limitation for an antioxidant substrate. Feedback regulation of synthesis and substrate availability can affect glutathione stores. [3,16]

Biological Mechanism

Glutathione (GSH) is a major intracellular antioxidant and redox buffer. It supplies reducing equivalents to glutathione peroxidases, which reduce hydrogen peroxide to water and organic hydroperoxides to their corresponding alcohols, producing oxidized glutathione (GSSG). Glutathione reductase uses NADPH to recycle GSSG to GSH. Glutathione S-transferases use GSH in conjugation reactions; glutathione also participates in protein-thiol redox regulation. [3] Cells synthesize GSH from its amino acid precursors, with feedback control at glutamate-cysteine ligase. [16] IV administration raises plasma GSH transiently and changes circulating thiol metabolites. [6,7] Uniform intact-GSH uptake into all tissues, including the brain, and equivalent SC exposure are separate extrapolations. Clinical outcomes therefore depend on route, formulation, tissue delivery, and the condition being treated. [5–11,18,19]

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